# Temporal Coded Spiking LCA

Alex Zhang

October 2026

## 1 Definitions

Let $u_i^{(k)}$ be the state of the integrator at cycle k.
Let $v$ be the state of the IF neuron attached to the integrator.
Let $t_r$ be the received spike time of the recurrent weights.
Let $t_f$ be the received spike time of the feedforward weights.
Let $\eta$ be the LCA learning rate.
Let $w_r$ be the recurrent weights leading to neuron $i$, with $w_r = \epsilon(I - \eta * \Phi^T\Phi)$.
Let $w_f$ be the feedforward weights leading to neuron $i$, with $w_f = \epsilon\eta\Phi^T$.
Let $T$ be the cycle length.
Let $\theta$ be the threshold for the attached IF neuron.
Let $a$ be the coefficients of sparse coding.
Let $x$ be the image vector.
Let $\lambda$ be the threshold/sparsity penalty coefficient.
Let $\epsilon$ be the decay constant of $u_i^{(k)}$.

## 2 Derivations

The LCA formulation for the state $u^{(k+1)}$ is the following:

$$u^{(k+1)} = (1-\epsilon)u^{(k)} + w_r a^{(k)} + w_f x$$

So, for each $u_i^{(k+1)}$,

$$u_i^{(k+1)} = (1-\epsilon)u_i^{(k)} + \sum_j x_j w_{f_j} + \sum_j a_j w_{r_j}$$

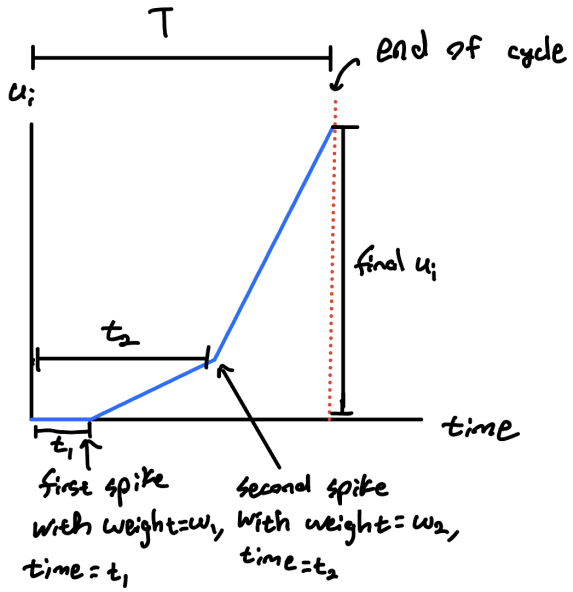
To begin, we must encode $a_j$ and $x_j$:

$$T - t_{r_j} = a_j$$
$$T - t_{f_j} = x_j$$

It then follows that the state of the next cycle $u_i^{(k+1)}$ is given by:

$$u_i^{(k+1)} = (1-\epsilon)u_i^{(k)} + \sum_j (T - t_{f_j}^{(k+1)})w_{f_j} + \sum_j (T - t_{r_j}^{(k+1)})w_{r_j}$$

This equation describes an integrator where weights are summed at spike times. The following example illustrates this regime:

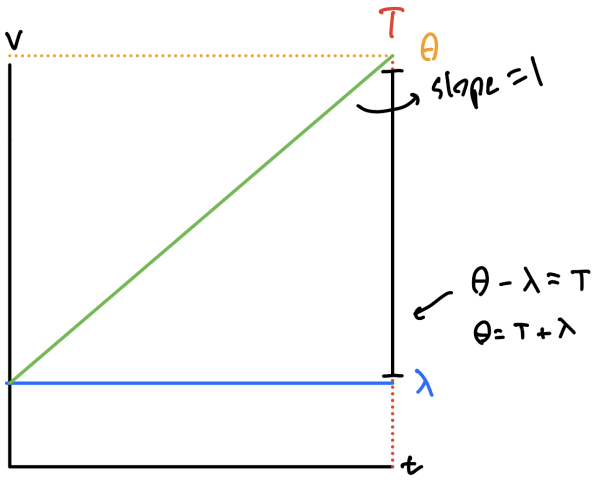



In this example, there are two spikes that come at times $t_1$ and $t_2$. The state $u_i$ begins at the previous cycle's final $u_i$ with a decay step applied (which for simplicity, we'll set to 0) and uses the two spikes' weights as input currents, meaning the slope from $t_1$ to $t_2$ is the weight $w_1$, and the slope from $t_2$ to the cycle end $T$ is $w_1 + w_2$. From here, one can derive the final $u_i$ in terms of the two spike times and the previous $u_i$ to be $u_i = (t_2 - t_1)w_1 + (T - t_2)(w_2 + w_1) = (T - t_1)w_1 + (T - t_2)w_2$ (you can see how $(t_2 - t_1)w_1 + (T - t_2)(w_2 + w_1)$ sum to the height of the final $u_i$ if you break the graph up into two triangles), which is equal to the aforementioned expression for $u_i^{(k)}$.

Furthermore, the next cycle's IF membrane state is determined by the following function on timestep $t$:

$$v_i^{(k+1)}(t) = u_i^{(k)} + t \text{ when } v_i < \theta \text{ and } 0 \text{ after } v_i \geq \theta$$

It has a constant input current of 1 and starts at $u_i^{(k)}$. The threshold $\theta$ is derived using the condition that the neuron should only fire if $u_i > \lambda$. The following diagram shows the derivation:




At the threshold, the constant input current should exactly reach the threshold at the end of the cycle, at time $T$. Thus, we can solve for $\theta$ in terms of $T$ and $\lambda$ by setting the distance $\theta - \lambda$ equal to the slope times distance $T$. As shown in the figure, $\theta = T + \lambda$.

From this, we can derive the output spike time $t^{(k+1)}$. The slope of 1 can be written as $1 = (\theta - u_i^{(k)})/t^{(k+1)} = (T + \lambda - u_i^{(k)})/t^{(k+1)}$, and solving for $t^{(k+1)}$, $t^{(k+1)} = (T + \lambda - u_i^{(k)})$. However, $t$ cannot be negative, so we must enforce positivity: $t^{(k+1)} = \max(0, (T - (u_i^{(k)} - \lambda))$. If there is a value $(u_i^{(k)} - \lambda) > T$, this value is too large for the representational range of the chosen cycle length T and input current of 1, and the maximum representable spike time (0) is applied. Of course, if $u_i + T < \theta = \lambda + T$, the neuron doesn't fire. If you plug back in the original $T - t_r = a_i$, you can see that, for the representable range, $a_i = u_i^{(k)} - \lambda$, matching LCA.